Postural Classification in Seated Position Using Machine Learning

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***Abstract*—This paper presents the development of a Machine Learning model aimed at classifying sitting postures, with the broader goal of promoting ergonomic practices and preventing musculoskeletal disorders in the workplace. The motivation behind this study lies in the growing concern about health problems related to prolonged incorrect postures during desk work. Data were collected using pressure sensors embedded in an ergonomic chair, capturing ten distinct sitting postures, each sustained for 30 seconds. Data analysis was performed on the Google Colab platform with the PyCaret library, which enabled efficient testing and comparison of multiple classification algorithms. The preliminary results indicate a high potential for accurate posture detection. A comparative evaluation of model performance and sensor data relevance is ongoing. The findings highlight the feasibility of integrating intelligent posture monitoring systems into everyday office environments, contributing to the advancement of ergonomics at work and proactive health monitoring.**

I. Introduction

Musculoskeletal disorders (MSDs) continue to pose a major challenge in occupational health, particularly in environments that require extended periods of sitting, such as control rooms, office spaces, and call centers. These conditions, often associated with sustained poor postural habits, not only impair individual well-being, but also reduce productivity and increase absenteeism [1] [2].

Recent developments in wearable technologies and sensor-based systems have opened new pathways to address this issue. Specifically, pressure-sensitive seating systems, when integrated with machine learning models, present a promising avenue for monitoring and classifying sitting postures in real time. Such technologies can play a crucial role in promoting early detection of harmful postural patterns and in guiding ergonomic interventions tailored to the user [3].

This study aims to design and implement a computational framework capable of automatically identifying various seated postures through data collected from pressure sensors. The ultimate goal is to support injury prevention strategies, improve ergonomic conditions in sedentary workplaces, and promote healthier occupational habits.

II. State of the art

The increasing incidence of work-related musculoskeletal disorders (MSDs), particularly in occupations requiring prolonged static postures, has driven interest in intelligent systems capable of

tracking and improving seated posture. Traditional ergonomic assessments, which rely on observational checklists or self-reports, are often subjective and lack real-time responsiveness, reducing their effectiveness in dynamic workplace settings.

A. Pressure Sensor-Based Posture Monitoring

Pressure sensors have proven to be an effective tool for posture monitoring, as they provide a non-intrusive, continuous method to measure pressure distribution over a seating surface. These sensors are often embedded in chairs or cushions and can detect subtle shifts in weight and position, enabling accurate identification of user posture [4]. Pressure data allows for spatial mapping of seated postures, which can be interpreted to classify different user positions over time.

B. Machine Learning in Posture Classification

Machine learning algorithms have become essential in interpreting pressure sensor data to identify sitting postures. Approaches such as k-nearest neighbors (KNN), support vector machines (SVM), and convolutional neural networks (CNNs) have demonstrated strong performance in classifying complex posture patterns. Achieving an accuracy rate above 99%, Tsai et al (2023) developed a posture recognition system capable of classifying ten postures using an ensemble learning approach. Similarly, Lee and Kim (2022) applied CNNs to children's posture data and successfully classified nine postures with high precision.

C. Practical Applications and Limitations

Recent reviews have consolidated findings from multiple posture monitoring systems, highlighting their utility in real-time feedback, ergonomic training, and long-term health management [5]. These systems are especially beneficial in office environments, where incorrect posture often persists unnoticed. However, challenges remain regarding generalization across individuals, calibration, and integration into standard workplace settings [6].

III. Metodology

This section outlines the experimental procedures, data acquisition strategy, and modeling approach used in this study. A systematic framework was adopted to ensure consistent data collection and evaluate the performance of machine learning algorithms in classifying seated postures based on input from the pressure sensor.

A. Data Acquisition

Following approval from the Ethics Committee of the Coimbra Polytechnic University, a total of 40 volunteers participated in the study, helping collect data using an instrumented chair equipped with 32x32 pressure sensors distributed throughout the seating surface (Fig. 1). These sensors were connected to a pressure mapping system, allowing real-time visualization and continuous recording of the data.

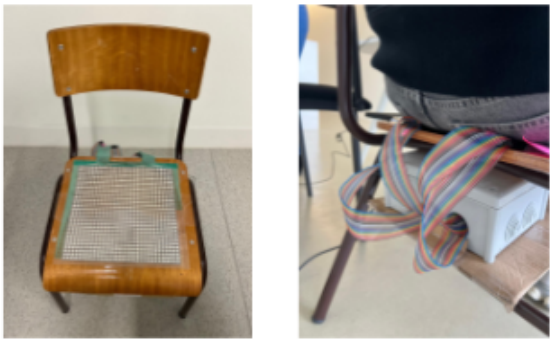


Fig. 1: Data aquisition setup. (a) Chair equipped with a matrix of 32x32 pressure sensors; (b) Participant during a data collection procedure.



Fig. 2: Postures used in data collection [7].

As it can be seen in Fig. 2, each volunteer was instructed to simulate 10 distinct sitting postures. Each posture was held for 30 seconds. The selected postures focus on the pressure-based ergonomic assessment for the seated configurations.

B. Preprocessing and Feature Extraction

To improve data quality, a median filter was applied to all pressure map images, reducing high-frequency noise while preserving relevant structural details. This helped counteract variations caused by sensor inconsistencies and small involuntary movements.

Beyond the image data, three features were extracted to enhance the model input: Body Mass Index (BMI), Center of Pressure (CoP), and Percentage of Pressure. These were tested individually and in combination with raw or filtered images to explore their impact on classification performance.

All configurations raw and filtered were retained for comparison during the evaluation phase.

C. Model Training and Evaluation

The classification pipeline was implemented using PyCaret in a Google Colab environment, enabling efficient preprocessing, model comparison, and reproducibility.

Five supervised learning algorithms were evaluated: Random Forest (RF), eXtreme Gradient Boosting (XGB), Support Vector Machine (SVM), Artificial Neural Networks (ANN), and k-Nearest Neighbors (KNN). Each was trained under various configurations combining raw or filtered pressure maps with feature sets such as BMI, Center of Pressure (CoP), and Percentage of Pressure.

All models were validated using 10-fold cross-validation to ensure generalization. Performance was measured using accuracy and F1-score, chosen for their ability to assess both overall correctness and balance between precision and recall in multi-class posture classification.

IV. Results

This section presents the performance of various machine learning models applied to posture classification using pressure map images and derived features.

A. Performance with Raw Data and BMI

1) Accuracy

The initial experiments were conducted using raw pressure map data across the five models. In the Table I, the ANN model achieved the highest baseline accuracy, with SVM and RF also showing strong initial results.

To evaluate the impact of the physiological context, Body Mass Index (BMI) was introduced as an external feature. Its inclusion led to a significant improvement in accuracy, especially in tree-based models. XGB, in particular, achieved the highest accuracy when used with filtered images and BMI. RF also benefited noticeably. SVM, however, suffered a steep performance drop, likely due to feature scale sensitivity and the non-linear effect of BMI.

TABLE I: Models Accuracy

Models	No Filter	Filter	No Filter + BMI	Filter + BMI
RF	0.9355	0.9343	0.9475	0.9391
XGB	0.9152	0.9140	0.9677	0.9761
SVM	0.9390	0.8866	0.2186	0.1637
ANN	0.9463	0.9379	0.9486	0.9427
KNN	0.5247	0.4780	0.7193	0.6727

2) F1-Score

F1-score was used as a complementary metric to account for imbalances between precision and recall. Here, too, ANN led the baseline evaluation with a score of 0.9449, confirming its ability to consistently detect posture classes. When BMI was introduced, XGB again achieved the best performance, reaching a score of 0.9750 with filtered data, closely followed by RF and ANN, as shown in Table II. The addition of BMI helped these models become more sensitive to postures influenced by weight distribution. In contrast, SVM's F1-score dropped significantly across all BMI-inclusive configurations, likely due to limitations in handling heterogeneous feature types without prior normalization.

TABLE II: Models F1-Score

Models	No Filter	Filter	No Filter + BMI	Filter + BMI
RF	0.9305	0.9282	0.9430	0.9332
XGB	0.9085	0.9079	0.9651	0.9750
SVM	0.9343	0.8781	0.1580	0.1149
ANN	0.9449	0.9328	0.9434	0.9371
KNN	0.5003	0.4559	0.6999	0.6551

B. Internal Variables: Center of Pressure and Percentage of Pressure

1) Accuracy

To assess the impact of internal spatial features, the center of pressure (CoP) and percentage of pressure were added to the model input, representing force balance and contact area, respectively.

The best accuracy was obtained by RF (0.9765) with raw data and Percentage of Pressure, followed by KNN (0.9699). ANN also performed well with CoP (0.9449 without and 0.9367 with filter). XGB remained consistent, though not surpassing its BMI-based results. SVM dropped sharply with Percentage of Pressure, especially without filter (0.0611), indicating sensitivity to noise.

2) F1-Score

F1-score results reinforced the value of the features. The best result was achieved by RF (0.9764) using raw data with Percentage of Pressure, showing strong balance between precision and recall.

ANN also performed well with CoP (0.9314 with filter) and filtered Percentage of Pressure (0.9299), indicating good adaptability to smoothed spatial inputs. In contrast, SVM dropped sharply, scoring just 0.0296 with raw Percentage of Pressure, confirming its sensitivity to noise.

KNN achieved a high score (0.9698) in the same raw configuration but underperformed elsewhere. XGB remained stable, peaking at 0.9536 with raw Percentage of Pressure.

V. Discussion

The results highlight how combining pressure map data with derived features can significantly enhance seated posture classification. While traditional models like RF and XGB consistently performed well, the inclusion of BMI and internal spatial variables such as CoP and Percentage of Pressure provided important performance boosts, particularly when paired with raw data.

ANN models showed robustness to different feature sets and benefited notably from filtered CoP data. Conversely, SVM exhibited limited adaptability, especially when exposed to unstructured inputs like % Pressure without preprocessing, reinforcing the importance of feature compatibility with model architecture.

Overall, tree-based models demonstrated better integration of heterogeneous data, whereas models like KNN were more sensitive to noise and filtering effects.

VI. Conclusion

This work presented a machine learning framework for automatic classification of sitting posture using pressure sensor data. Through extensive testing across multiple algorithms and feature configurations, we demonstrated that enriching raw image data with biometric and spatial features, can significantly improve classification performance.

The best results were achieved using XGB and ANN models with minimal filtering, confirming their suitability for real-time ergonomic applications. Future work will explore real-time deployment, temporal posture tracking, and the inclusion of dynamic features for broader ergonomic monitoring.

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